

Chidaksh Ravuru

Research Interests

Whether the signal that certifies a system actually tracks the behavior you care about: reward and verifier design, trace-level analysis of how agents fail, and adversarial evaluation that tries to break a result before anyone trusts it. Most of what I have found came from reading the data by hand, annotating trajectories, auditing graders, and tracing a surprising number back to its cause.

Education

- 2024–2026 **University of North Carolina at Chapel Hill**, MS in Computer Science (Graduated)
Advisor: Prof. Shashank Srivastava, GPA: 4.0/4.0
Relevant Courses: Reinforcement Learning (RLHF/PPO), Reliable ML, Advanced LLM Agents, Bayesian Nash Equilibrium
- 2020–2024 **Indian Institute of Technology, Dharwad**, B.Tech in Computer Science
Advisors: Prof. Prabuchandran K.J. & Prof. Rajshekhar Bhat, GPA: 9.24/10

Publications

- 2026 **UNMASK: Discovering and Causally Verifying Spurious Shortcuts in Text Classifiers**, *COLM 2026*
Chidaksh Ravuru, Shashank Srivastava
- 2026 **Behavioral Coherence: A Method for Sensitive-Domain LLM Evaluation**, *Under Review, CHI 2027*
Anika Sharma, Malavika Mampally, **Chidaksh Ravuru**, Kandyce Brennan, Neil Gaikwad
- 2024 **Agentic Retrieval-Augmented Generation for Time Series Analysis**, *KDD 2024*
Chidaksh Ravuru, Sagar Sakhinana, Venkataramana Runkana
- 2024 **RESTORE: Graph Embedding Assessment Through Reconstruction**, *Preprint*
Hong Yung Yip, **Chidaksh Ravuru**, Neelabha Banerjee, Shashwat Jha, Amit Sheth, Aman Chadha, Amitava Das

Workshop Papers

- 2024 **Parameter-Efficient Quantized Mixture-of-Experts Meets Vision-Language Instruction Tuning**, *ICML, Workshop on Foundational Models*
Sagar Sakhinana, **Chidaksh Ravuru**, Geethan Sannidhi, Venkataramana Runkana

Research Experience

- Jun 2026–Present **AI Engineer**, *Excipy LLC*, Remote
- Audited a hand-typed pharmacogenomic evidence table against live **PharmGKB** and found **6 of 17 rows** claiming grades the source did not support; replaced it with **334 generated rows across 78 drugs**, each tied to its source annotation, and added a **second-vendor model** that verifies every extracted non-response rate.
 - Designing the platform's **backend**: every pipeline run stored in **S3** by drug and configuration so a repeat search becomes a lookup, and companies, licences and approval rules in one **reference database** that a person approves before each scheduled refresh, with alerts when a rule's quoted legal text changes.
- Summer 2025 **Applied Scientist Intern**, *Pavo AI*, San Francisco, CA
- Trained **persona-conditioned Process Reward Models** (PyTorch, HuggingFace) to guide search-space exploration in a planning agent, conditioning on correctness *and* persona fit; shipped to production and improved task performance by **10%**.
 - Built **PersonaBench**, **3,000 scored trajectories**, through a student-teacher pipeline in which the teacher verified every scraped source and flagged hallucinations before a trace entered the training set, keeping fabricated context out of the data the model trained on.
- Summer 2023 **Machine Learning Research Intern**, *Tata Research, Development and Design Centre*, Bangalore, India
- Developed an agentic RAG pipeline with structured **data curation**, **DPO**-based post-training, and multi-step retrieval orchestration, achieving a **12% improvement** in prediction accuracy across temporal forecasting tasks, contributing to the **KDD 2024 publication**.
 - Instrumented an **automated evaluation pipeline** across **diverse task benchmarks**, measuring model capability, identifying failure modes, and driving **18% performance gains** through targeted fine-tuning, establishing a feedback loop between evaluation findings and model improvement.

Selected Research

- 2026 **UNMASK: Causal Verification of Spurious Shortcuts**, [COLM 2026](#)
- Automated pipeline that writes candidate shortcuts as executable boolean functions, filters them through statistical validation with independent replication and **FDR control**, and credits one as a real driver only after a **counterfactual intervention** confirms the model relies on it; verified 9 of 10 features on BERT and improved **HANS accuracy +12.58 pp** while keeping MNLI-matched within 1.2 pp.
 - Used the verified features as annotation-free groups for DFR, matching hand-labeled worst-group accuracy on **CivilComments** (71.84 vs. 70.1) without demographic labels. Run unmodified on **RewardBench2**, it found Safety preferences tracking refusal phrasing (**odds ratio 161** on personal-info prompts) and length sensitivity in the label-neutral Ties subset.
- 2026 **Does Structure Survive Scale? Diagnosing Hierarchy in SAEs**, *EleutherAI SOAR, ongoing*
- Building a coverage-based diagnostic that tests whether hierarchy-recovery methods (**Matryoshka SAEs**, **Temporal SAEs**) produce coherent parent-child feature structure on **Gemma-2-2B**, benchmarked against **Gemma Scope**; in a 6-person EleutherAI pod advised by **Gonalo Paulo**.
 - Trained SAEs on PCFG transformer activations and designed controlled grammars to isolate *which* properties of natural language make these methods fail.
- 2026 **Unlearning Robustness: Can Cheap Signals Predict Relearning?**, *Independent*
- **Pre-registered** a test of whether cheap, attack-free signals on an unlearned checkpoint predict how much knowledge an attack recovers, across **8 methods and 32 OpenUnlearning checkpoints** of Llama-3.2-1B on TOFU; quantization and direction ablation recovered nothing at this scale, so the test ran on relearning.
 - **No signal predicted recovery across methods**: a $\rho = +0.90$ on 5 checkpoints fell to -0.35 on 27, and leave-one-method-out was at chance (3/8). Weight distance's $\rho = -0.85$ came from within each method's hyperparameter sweep.
- 2026 **Trace-Level Failure Taxonomy for Tool-Using Agents**, [ToolBench](#)
- Hand-annotated **500 agent execution traces** at step level, separating tool-selection errors, argument errors, and hallucinated API calls into a **behavioral taxonomy** rather than a single pass/fail score.
 - Turned the taxonomy into a preference dataset and **DPO-trained** Llama-3.1-8B on it, improving tool-selection accuracy **9%** and cutting hallucinated API calls **22%** on a **held-out** set of 100 tasks.
- 2026 **Grader Telemetry for LLM-as-Judge Evaluation**, [Code](#) | [Dashboard](#)
- Built the observability layer a judge needs before its scores are trusted: **per-element attribution**, judge self-consistency via **Krippendorff's alpha**, and **position-bias detection**, surfaced in a React dashboard instead of buried in a results table.
 - Wrote the statistics underneath: **paired-bootstrap** hierarchical resampling, **Cohen's d**, and a minimum-detectable-effect test that reports a gap inside the noise floor as a **close call rather than a winner**.
- 2026 **Tracing a False Win in an Automated Optimization Loop**, *Independent*
- Integrated **GEPA**, a reflective prompt optimizer, with **Inspect AI's** agent evaluation harness at GEPA's **native extension seam**, reading unfamiliar source rather than wrapping the public API, and never mocking the real eval-scorer boundary.
 - A validation-selected candidate beat the seed prompt and then regressed on held-out data; **per-sample analysis** traced it to an undersized validation set, and a rerun at larger size rejected every overfit candidate and kept the seed. Reported both runs, including the one that failed.

Skills

Safety & Evals	Reward & Verifier Auditing, Trace Analysis, Failure Taxonomies, Adversarial Robustness & Held-Out Evaluation, Shortcut Auditing, LLM-as-Judge Validation, Evaluation Design
Interpretability	Sparse Autoencoders (Matryoshka, Temporal, Gemma Scope), Probes, Activation-Level Interventions (steering, direction ablation), Weight-Space Diagnostics, TransformerLens
Methods	Causal Verification (Counterfactual Intervention), Post-Training (DPO, Reward Modeling, PRM), Statistical Analysis (Bootstrap CIs, Effect Sizes, FDR Control, Inter-Rater Reliability)
Engineering	Python, PyTorch, JAX, C++, TypeScript, HuggingFace Transformers/TRL, PEFT/LoRA, vLLM, Inspect AI, React, FastAPI, Slurm, Docker, AWS, Weights & Biases

Teaching & Achievements

- Summer 2025 **Summer School Teacher**, *UNC School of Medicine*
Led a lecture series for high school students on Statistics and Machine Learning (Linear Regression, SVMs) in neuroscience and medical imagery.
- 2025–2026 **Graduate Teaching Assistant**, *UNC School of Data Science and Society*
TA for DATA 110: Intro to Data Science. Conducted lab recitations and taught basic probability, statistics, and Python coding for undergraduates.
- Reviewing **ACL, COLM, AIES**
- Fellowships **EleutherAI SOAR Fellow**, advised by **Gonalo Paulo** · **BlueDot Impact**: AI Safety Technical Course, Frontier AI Governance Course
- Awards Top 1% for **Google Research Week** (India, 2024) · Fully funded at **MLSS 2022** (Krakow) · **4th place** Inter IIT Techfest 2023